Aircraft vibration - guided synthetic demonstration

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1. Executive assessment

Verdict: The evidence supports a BPFO (Outer Race Fault) in the rolling bearing on the driving pinion shaft, based on 5 matched harmonic components detected in both flight and ground check recordings.

Provisional engineering assessment from the recorded evidence; the physical cause still requires confirmation by inspection.

Selected explanation: Outer-race bearing fault - supported. [E17] [E18] [E19] [E20]

2. Scope and operating conditions

Operator-planned synthetic demonstration with operator report review. Flight and ground are separate recordings; analyses use source-local clocks. No cross-source alignment or temperature correction was performed. Applicable references are AIR-SETUP-R2 and AIR-QUALITY-R2, both revision R2.

Reference documents

- quality.md (revision R2; source SHA256 b091bcef64337bd9)
- machine.json (revision R2; source SHA256 aceaaa70ccaec892)
- setup.md (revision R2; source SHA256 95b5456a15946872)

3. Data quality and preparation

Complete-source native profiles checked finite values and timestamp gaps. These checks do not certify sensor ranges or clipping. Original acceleration was retained. RAG supplied reference metadata; measurements came from native analysis tools.

Recording quality summary

Table 1: Recording quality summary

Source	Channels	Rows	Start row	Non-finite	Timestamp gaps	Evidence
mission_computer.csv	650	960	0	0 / 650	0 / 650	[E02]
flight_daq.mpai	24	614,400	0	0 / 24	0 / 24	[E03]
ground_check.mpai	24	614,400	0	0 / 24	0 / 24	[E04]
mission_computer.csv	3	160	39	0 / 3	0 / 3	[E06]
mission_computer.csv	3	160	639	0 / 3	0 / 3	[E08]
flight_daq.mpai	2	614,400	0	0 / 2	0 / 2	[E11]
ground_check.mpai	2	614,400	0	0 / 2	0 / 2	[E12]
mission_computer.csv	3	960	0	0 / 3	0 / 3	[E21]
ground_check.mpai	24	614,400	0	0 / 24	0 / 24	[E22]
mission_computer.csv	650	960	0	0 / 650	0 / 650	[E23]
flight_daq.mpai	24	614,400	0	0 / 24	0 / 24	[E24]
mission_computer.csv	3	960	0	0 / 3	0 / 3	[E25]
mission_computer.csv	3	160	639	0 / 3	0 / 3	[E26]
mission_computer.csv	3	160	39	0 / 3	0 / 3	[E27]
mission_computer.csv	650	960	0	0 / 650	0 / 650	[E28]

Quality counts show affected / assessed channels. Full channel profiles are available in the connected results and evidence companion. The channel profiles do not assess clipping against declared sensor ranges.

4. Measured findings

In the requested early (2-10 s) and late (32-40 s) controller windows, GBX_MONITOR_RMS means rise from 0.112330 to 0.401811 g. Speed stays at 1200 rpm and torque means are 60.0066 and 60.0185 percent. Full-record sensor_a/ sensor_b RMS values are 0.171174/0.119389 g in flight and 0.171149/0.119371 g on the ground. Four complete-record bearing analyses support the approximately 107.4 Hz BPFO family with five matched harmonics each. Measured-encoder order tracking was performed only on flight sensor_a; it is rotational context, not a bearing verdict. The FFT plots provide a separate comparison of the selected flight windows.

Measured-encoder order analysis coverage

Table 2: Measured-encoder order analysis coverage

Recording	Physical encoder	Vibration channel	Rows	Evidence
flight_daq.mpai	keyphasor	sensor_a	614,400	[E10]

Requested channel measurements

Table 3: Requested channel measurements

Source	Channel	Unit	RMS	Sample std	Valid / total
S02 [E02]	GBX_MONITOR_RMS		0.342855	0.222387	960 / 960
S03 [E03]	sensor_a	g	0.171174	0.170107	614400 / 614400
S03 [E03]	sensor_b	g	0.119389	0.119358	614400 / 614400
S04 [E04]	sensor_a	g	0.171149	0.170074	614400 / 614400
S04 [E04]	sensor_b	g	0.119371	0.119339	614400 / 614400
S02 [E06]	GBX_MONITOR_RMS		0.197916	0.163461	160 / 160
S02 [E08]	GBX_MONITOR_RMS		0.423474	0.134131	160 / 160
S03 [E11]	sensor_a	g	0.171174	0.170107	614400 / 614400
S03 [E11]	sensor_b	g	0.119389	0.119358	614400 / 614400
S04 [E12]	sensor_a	g	0.171149	0.170074	614400 / 614400
S04 [E12]	sensor_b	g	0.119371	0.119339	614400 / 614400
S02 [E21]	GBX_MONITOR_RMS		0.342855	0.222387	960 / 960
S04 [E22]	sensor_a	g	0.171149	0.170074	614400 / 614400
S04 [E22]	sensor_b	g	0.119371	0.119339	614400 / 614400
S02 [E23]	GBX_MONITOR_RMS		0.342855	0.222387	960 / 960
S03 [E24]	sensor_a	g	0.171174	0.170107	614400 / 614400
S03 [E24]	sensor_b	g	0.119389	0.119358	614400 / 614400
S02 [E25]	GBX_MONITOR_RMS		0.342855	0.222387	960 / 960
S02 [E26]	GBX_MONITOR_RMS		0.423474	0.134131	160 / 160
S02 [E27]	GBX_MONITOR_RMS		0.197916	0.163461	160 / 160
S02 [E28]	GBX_MONITOR_RMS		0.342855	0.222387	960 / 960

Retained cached scalar rows for explicitly requested channels and derived outputs; other channels remain in the full native review. [E02]: 960 samples, starting at row 0; [E03]: 614400 samples, starting at row 0; [E04]: 614400 samples, starting at row 0; [E06]: 160 samples, starting at row 39; [E08]: 160 samples, starting at row 639; [E11]: 614400 samples, starting at row 0; [E12]: 614400 samples, starting at row 0; [E21]: 960 samples, starting at row 0; [E22]: 614400 samples, starting at row 0; [E23]: 960 samples, starting at row 0; [E24]: 614400 samples, starting at row 0; [E25]: 960 samples, starting at row 0; [E26]: 160 samples, starting at row 639; [E27]: 160 samples, starting at row 39; [E28]: 960 samples, starting at row 0. Separate-source statistics do not establish time alignment or matching operating conditions. Use the documented setup and intervention before interpreting differences.

Measurement overview - Bearing & Gear Fault Diagnosis / flight_daq.mpai / sensor_a [E13]

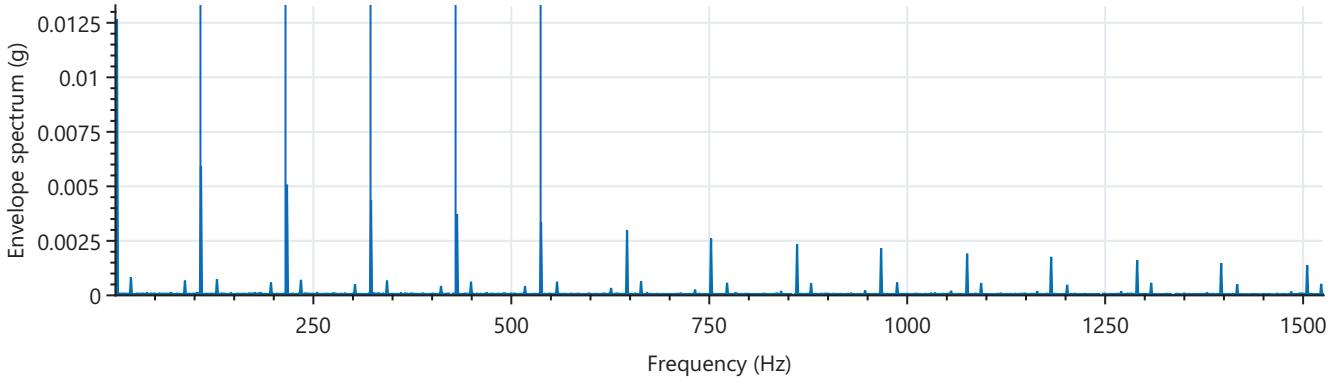


Figure 1: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-1525.98 Hz; full source selection retained. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz; Measured BPFO 5x at 537.012 Hz. AC spectrum; DC and its adjacent window bin are outside this display viewport. The complete analysis is retained.

Bearing-frequency detail - Bearing & Gear Fault Diagnosis / flight_daq.mpai / sensor_a [E13]

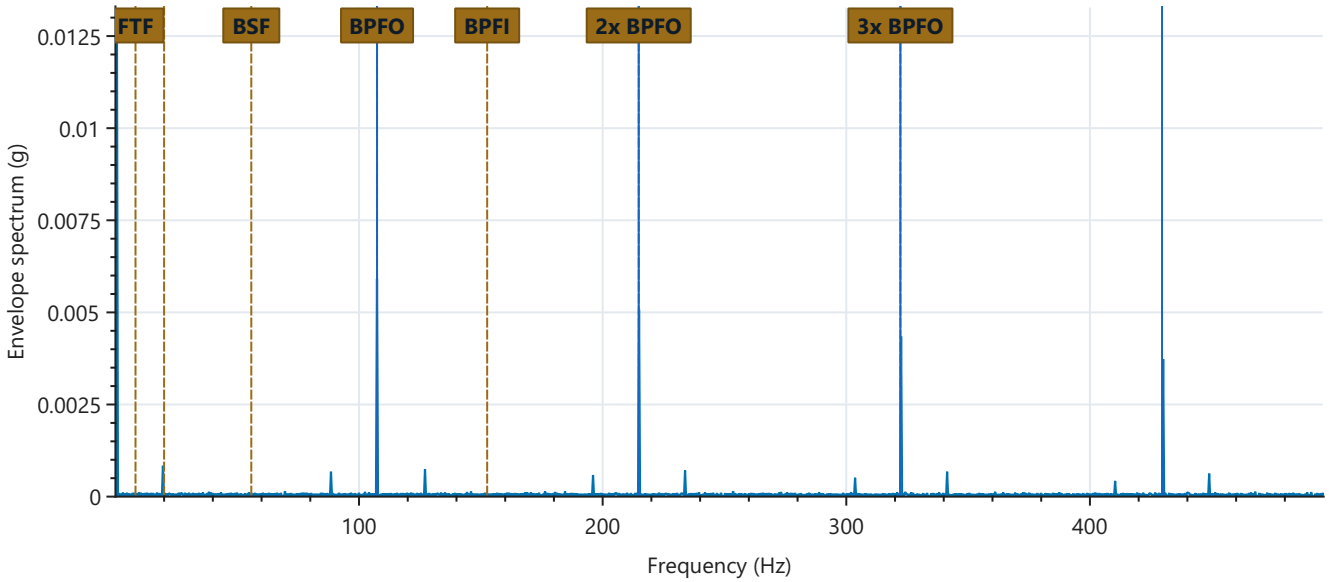


Figure 2: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-495.945 Hz; full source selection retained. Reference markers (dashed amber; not detected peaks): BPFI at 152.598 Hz; BPFO at 107.402 Hz; BSF at 55.788 Hz; FTF at 8.26167 Hz; 1x shaft at 20 Hz; 2x BPFO at 214.803 Hz; 3x BPFO at 322.205 Hz. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz. Read alongside the overview. Reference coordinates come from the current native result; a marker is not evidence of a detected defect.

Processing and operating conditions [E13]

Table 4: Processing and operating conditions [E13]

Parameter	Value
Sample rate (Hz)	12,800
Envelope band (Hz)	3200 to 6400
Envelope FFT size	1,048,576
Frequency resolution (Hz)	0.012207
Window	Hann (envelope domain)
Averaging	Arithmetic mean of segment magnitudes
Segments	1
Segment overlap fraction	0.5
Amplitude convention	One-sided peak amplitude, Hann coherent-gain corrected
Speed (rpm)	1,200
Speed standard deviation (rpm)	0
Speed reference	measured_channel_mean
Speed channel	shaft_rpm
Rolling elements	13
Pitch diameter (mm)	60
Element diameter (mm)	10.43
Contact angle (deg)	0

Differential signature checks [E13]

Table 5: Differential signature checks [E13]

Family	Reference (Hz)	Result	Unique harmonics	Sidebands	Mean prominence ratio
BPFO	107.402	supported	5	0	84.0166
BPFI	152.598	insufficient	1	0	20.4598
BSF	55.788	insufficient	0	0	0
FTF	8.26167	insufficient	0	0	0

Insufficient means the tested spectrum lacks qualifying evidence; it does not establish absence. Rejected means a recorded explanation has explicit counterevidence and a reason. Prominence is amplitude divided by the native local median noise floor, not a probability or calibrated SNR. Sideband counts exclude carriers discarded for overlap; raw modulation may remain unqualified.

Measured harmonic lines [E13]

Table 6: Measured harmonic lines [E13]

Family	Harmonic	Measured (Hz)	Deviation (Hz)	Peak amplitude (g)	Prominence ratio
BPFO	1	107.397	-0.00420573	0.00591496	78.5927
BPFO	2	214.807	0.00379557	0.00507472	85.3106
BPFO	3	322.205	-0.000410156	0.00434819	91.0949
BPFO	4	429.602	-0.00461589	0.00373297	71.7894
BPFO	5	537.012	0.00338542	0.00334364	93.2953

Measurement overview - Bearing & Gear Fault Diagnosis / flight_daq.mpai / sensor_b [E14]

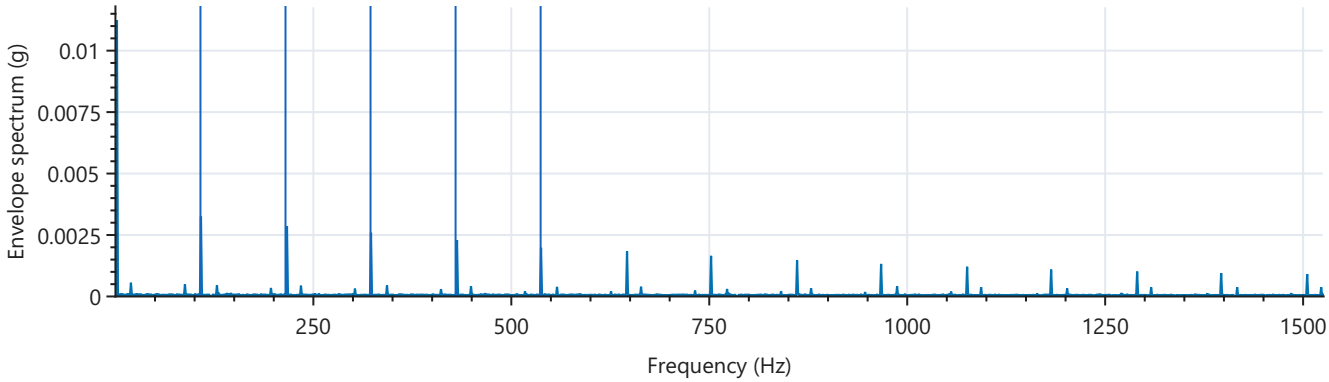


Figure 3: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-1525.98 Hz; full source selection retained. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz; Measured BPFO 5x at 537.012 Hz. AC spectrum; DC and its adjacent window bin are outside this display viewport. The complete analysis is retained.

Bearing-frequency detail - Bearing & Gear Fault Diagnosis / flight_daq.mpai / sensor_b [E14]

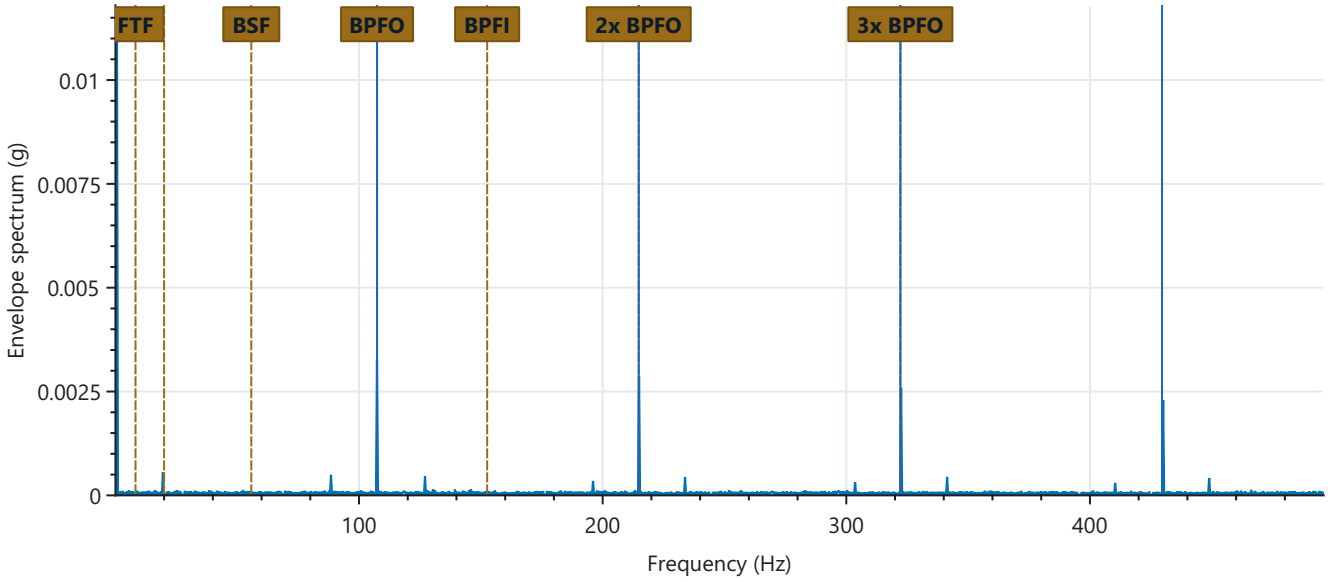


Figure 4: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-495.945 Hz; full source selection retained. Reference markers (dashed amber; not detected peaks): BPFI at 152.598 Hz; BPFO at 107.402 Hz; BSF at 55.788 Hz; FTF at 8.26167 Hz; 1x shaft at 20 Hz; 2x BPFO at 214.803 Hz; 3x BPFO at 322.205 Hz. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz. Read alongside the overview. Reference coordinates come from the current native result; a marker is not evidence of a detected defect.

Processing and operating conditions [E14]

Table 7: Processing and operating conditions [E14]

Parameter	Value
Sample rate (Hz)	12,800
Envelope band (Hz)	3200 to 6400
Envelope FFT size	1,048,576
Frequency resolution (Hz)	0.012207
Window	Hann (envelope domain)
Averaging	Arithmetic mean of segment magnitudes
Segments	1
Segment overlap fraction	0.5
Amplitude convention	One-sided peak amplitude, Hann coherent-gain corrected
Speed (rpm)	1,200
Speed standard deviation (rpm)	0
Speed reference	measured_channel_mean
Speed channel	shaft_rpm
Rolling elements	13
Pitch diameter (mm)	60
Element diameter (mm)	10.43
Contact angle (deg)	0

Differential signature checks [E14]

Table 8: Differential signature checks [E14]

Family	Reference (Hz)	Result	Unique harmonics	Sidebands	Mean prominence ratio
BPFO	107.402	supported	5	0	53.3397
BPFI	152.598	insufficient	1	0	60.2573
BSF	55.788	insufficient	0	0	0
FTF	8.26167	insufficient	0	0	0

Insufficient means the tested spectrum lacks qualifying evidence; it does not establish absence. Rejected means a recorded explanation has explicit counterevidence and a reason. Prominence is amplitude divided by the native local median noise floor, not a probability or calibrated SNR. Sideband counts exclude carriers discarded for overlap; raw modulation may remain unqualified.

Measured harmonic lines [E14]

Table 9: Measured harmonic lines [E14]

Family	Harmonic	Measured (Hz)	Deviation (Hz)	Peak amplitude (g)	Prominence ratio
BPFO	1	107.397	-0.00420573	0.0032501	50.3014
BPFO	2	214.807	0.00379557	0.00286072	70.4284
BPFO	3	322.205	-0.000410156	0.0025937	61.5546
BPFO	4	429.602	-0.00461589	0.00229591	50.4669
BPFO	5	537.012	0.00338542	0.00198093	33.9474

Measurement overview - Bearing & Gear Fault Diagnosis / ground_check.mpai / sensor_a [E15]

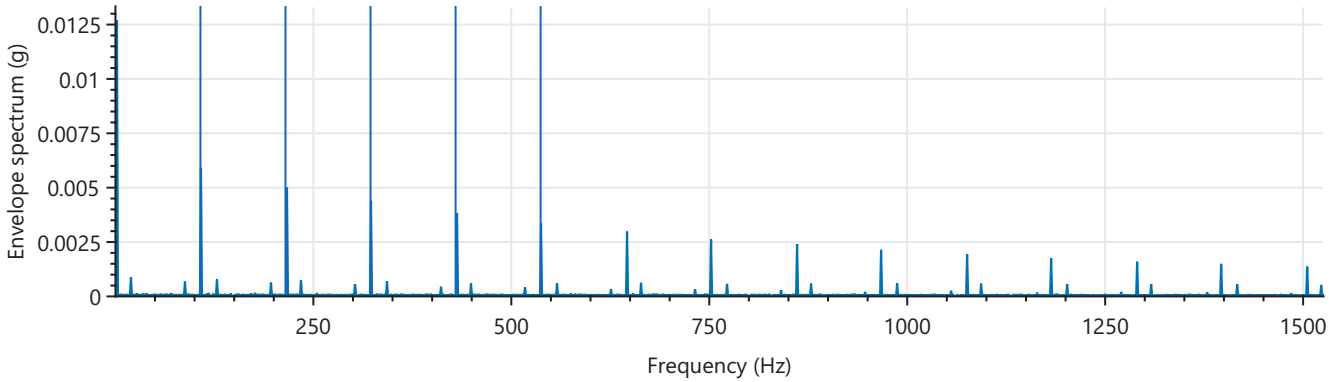


Figure 5: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-1525.98 Hz; full source selection retained. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz; Measured BPFO 5x at 537.012 Hz. AC spectrum; DC and its adjacent window bin are outside this display viewport. The complete analysis is retained.

Bearing-frequency detail - Bearing & Gear Fault Diagnosis / ground_check.mpai / sensor_a [E15]

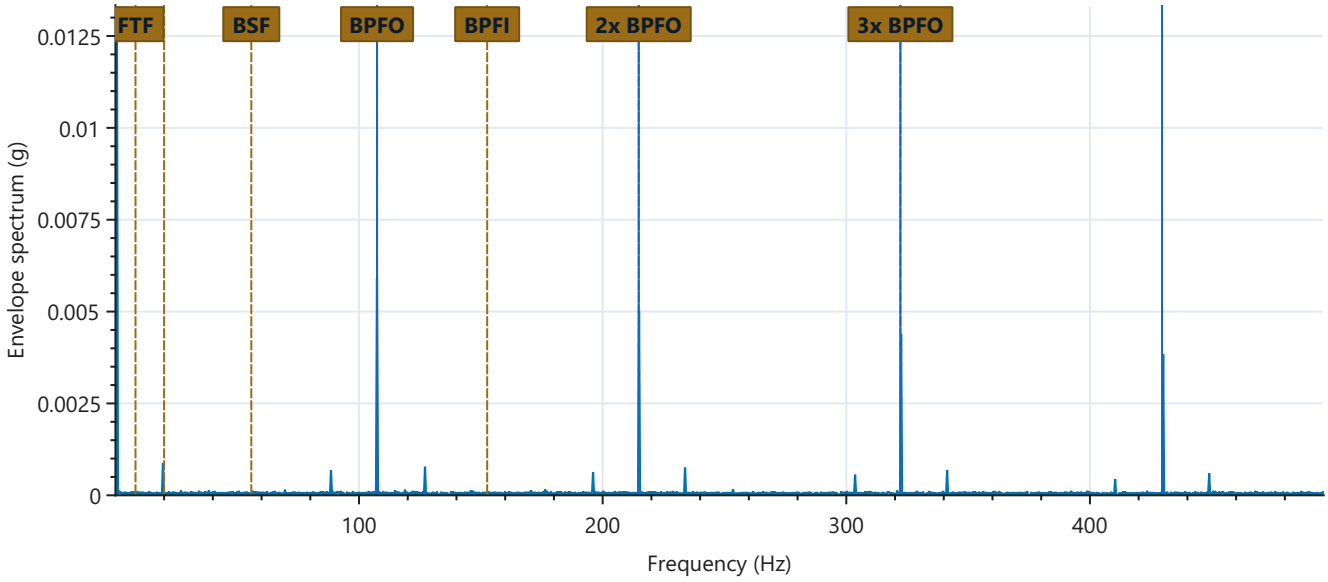


Figure 6: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-495.945 Hz; full source selection retained. Reference markers (dashed amber; not detected peaks): BPFI at 152.598 Hz; BPFO at 107.402 Hz; BSF at 55.788 Hz; FTF at 8.26167 Hz; 1x shaft at 20 Hz; 2x BPFO at 214.803 Hz; 3x BPFO at 322.205 Hz. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz. Read alongside the overview. Reference coordinates come from the current native result; a marker is not evidence of a detected defect.

Processing and operating conditions [E15]

Table 10: Processing and operating conditions [E15]

Parameter	Value
Sample rate (Hz)	12,800
Envelope band (Hz)	3200 to 6400
Envelope FFT size	1,048,576
Frequency resolution (Hz)	0.012207
Window	Hann (envelope domain)
Averaging	Arithmetic mean of segment magnitudes
Segments	1
Segment overlap fraction	0.5
Amplitude convention	One-sided peak amplitude, Hann coherent-gain corrected
Speed (rpm)	1,200
Speed standard deviation (rpm)	0
Speed reference	measured_channel_mean
Speed channel	shaft_rpm
Rolling elements	13
Pitch diameter (mm)	60
Element diameter (mm)	10.43
Contact angle (deg)	0

Differential signature checks [E15]

Table 11: Differential signature checks [E15]

Family	Reference (Hz)	Result	Unique harmonics	Sidebands	Mean prominence ratio
BPFO	107.402	supported	5	0	89.4179
BPFI	152.598	insufficient	1	0	16.517
BSF	55.788	insufficient	0	0	0
FTF	8.26167	insufficient	0	0	0

Insufficient means the tested spectrum lacks qualifying evidence; it does not establish absence. Rejected means a recorded explanation has explicit counterevidence and a reason. Prominence is amplitude divided by the native local median noise floor, not a probability or calibrated SNR. Sideband counts exclude carriers discarded for overlap; raw modulation may remain unqualified.

Measured harmonic lines [E15]

Table 12: Measured harmonic lines [E15]

Family	Harmonic	Measured (Hz)	Deviation (Hz)	Peak amplitude (g)	Prominence ratio
BPFO	1	107.397	-0.00420573	0.00591292	126.004
BPFO	2	214.807	0.00379557	0.00500364	90.6273
BPFO	3	322.205	-0.000410156	0.00439597	65.1173
BPFO	4	429.602	-0.00461589	0.00383717	75.9809
BPFO	5	537.012	0.00338542	0.00336717	89.3601

Measurement overview - Bearing & Gear Fault Diagnosis / ground_check.mpai / sensor_b [E16]

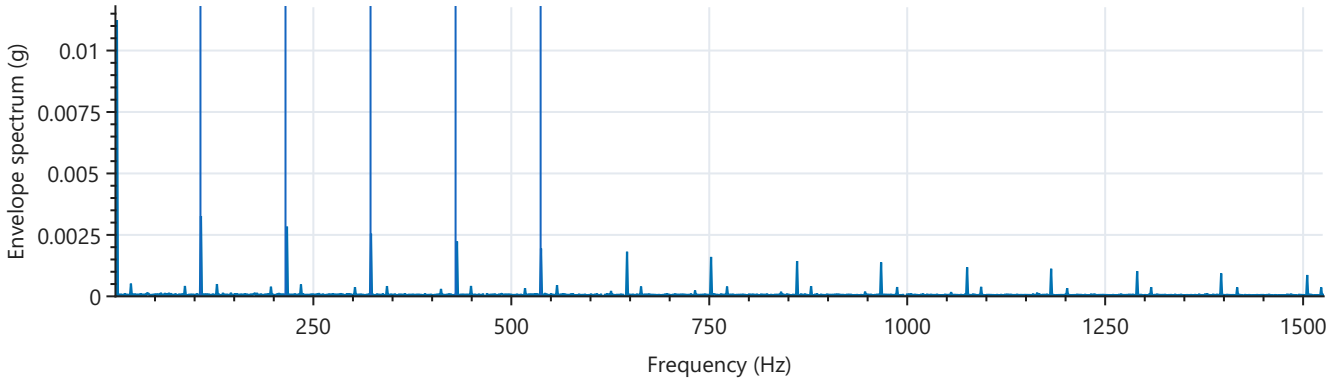


Figure 7: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-1525.98 Hz; full source selection retained. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz; Measured BPFO 5x at 537.012 Hz. AC spectrum; DC and its adjacent window bin are outside this display viewport. The complete analysis is retained.

Bearing-frequency detail - Bearing & Gear Fault Diagnosis / ground_check.mpai / sensor_b [E16]

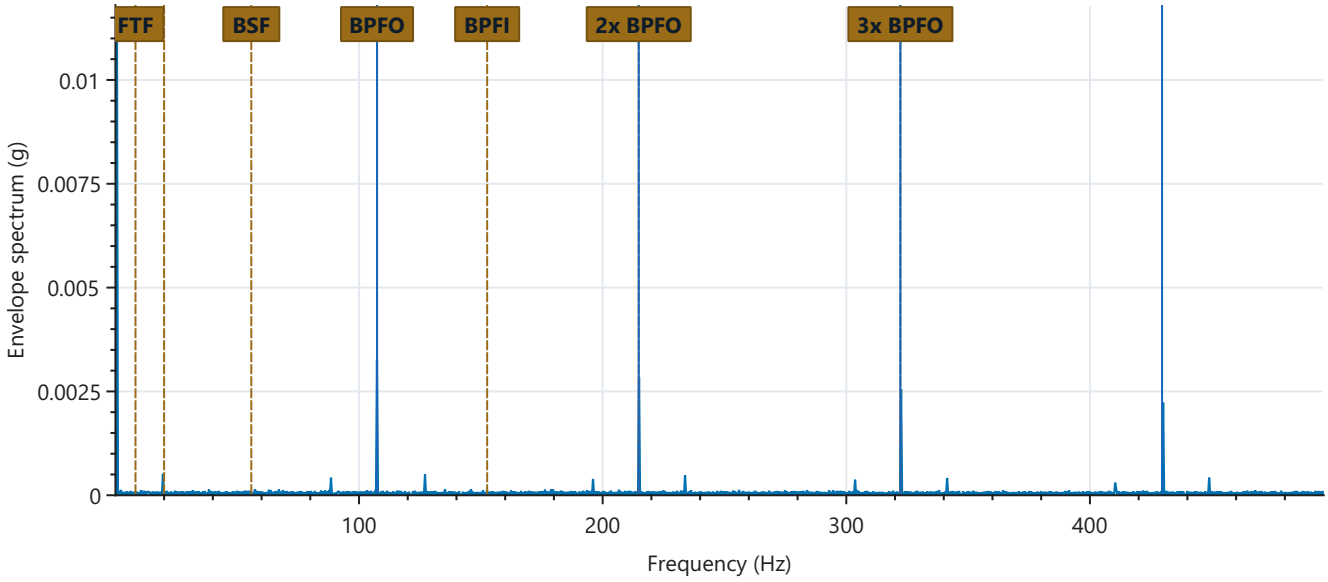


Figure 8: Source: Bearing & Gear Fault Diagnosis · 1 channel · complete record analysed (Min-Max display reduction). Display viewport: 0.0244141-495.945 Hz; full source selection retained. Reference markers (dashed amber; not detected peaks): BPFI at 152.598 Hz; BPFO at 107.402 Hz; BSF at 55.788 Hz; FTF at 8.26167 Hz; 1x shaft at 20 Hz; 2x BPFO at 214.803 Hz; 3x BPFO at 322.205 Hz. Measured qualifying lines (solid blue): Measured BPFO 1x at 107.397 Hz; Measured BPFO 2x at 214.807 Hz; Measured BPFO 3x at 322.205 Hz; Measured BPFO 4x at 429.602 Hz. Read alongside the overview. Reference coordinates come from the current native result; a marker is not evidence of a detected defect.

Processing and operating conditions [E16]

Table 13: Processing and operating conditions [E16]

Parameter	Value
Sample rate (Hz)	12,800
Envelope band (Hz)	3200 to 6400
Envelope FFT size	1,048,576
Frequency resolution (Hz)	0.012207
Window	Hann (envelope domain)
Averaging	Arithmetic mean of segment magnitudes
Segments	1
Segment overlap fraction	0.5
Amplitude convention	One-sided peak amplitude, Hann coherent-gain corrected
Speed (rpm)	1,200
Speed standard deviation (rpm)	0
Speed reference	measured_channel_mean
Speed channel	shaft_rpm
Rolling elements	13
Pitch diameter (mm)	60
Element diameter (mm)	10.43
Contact angle (deg)	0

Differential signature checks [E16]

Table 14: Differential signature checks [E16]

Family	Reference (Hz)	Result	Unique harmonics	Sidebands	Mean prominence ratio
BPFO	107.402	supported	5	0	56.8554
BPFI	152.598	insufficient	1	0	44.1449
BSF	55.788	insufficient	0	0	0
FTF	8.26167	insufficient	0	0	0

Insufficient means the tested spectrum lacks qualifying evidence; it does not establish absence. Rejected means a recorded explanation has explicit counterevidence and a reason. Prominence is amplitude divided by the native local median noise floor, not a probability or calibrated SNR. Sideband counts exclude carriers discarded for overlap; raw modulation may remain unqualified.

Measured harmonic lines [E16]

Table 15: Measured harmonic lines [E16]

Family	Harmonic	Measured (Hz)	Deviation (Hz)	Peak amplitude (g)	Prominence ratio
BPFO	1	107.397	-0.00420573	0.00326532	79.6171
BPFO	2	214.807	0.00379557	0.00283476	60.2472
BPFO	3	322.205	-0.000410156	0.0025509	62.7468
BPFO	4	429.602	-0.00461589	0.00222979	35.1112
BPFO	5	537.012	0.00338542	0.00194593	46.5547

Measurement overview - FFT Spectrum Analysis / flight_daq.mpai [E07]

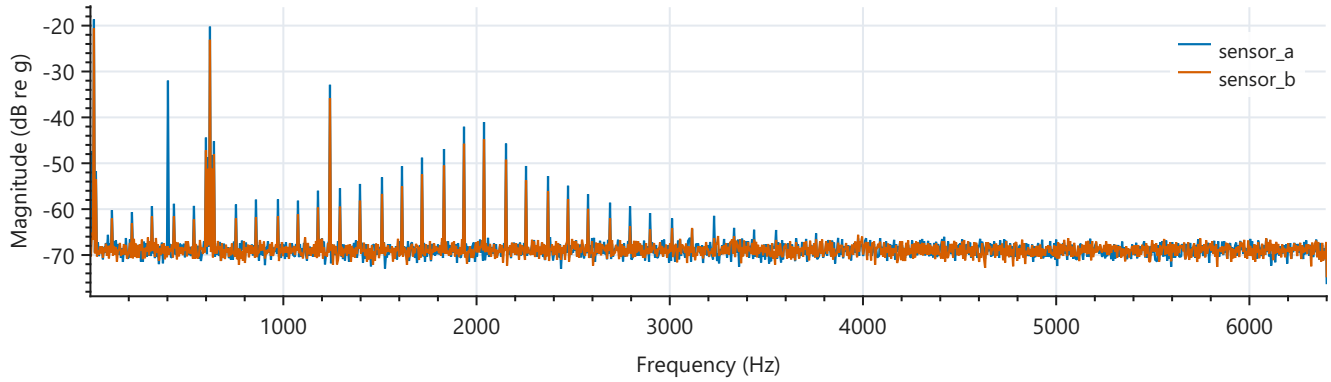


Figure 9: Source: FFT Spectrum Analysis · 2 channels · analysed 102,401 of 614,400 samples from sample 25,600 (Min-Max display reduction). dB amplitude from the cached native spectrum; all selected channels use the same amplitude convention.

Processing and operating conditions [E07]

Table 16: Processing and operating conditions [E07]

Parameter	Value
Sample rate (Hz)	12,800
FFT frequency resolution (Hz)	1.5625
FFT averages used	25
FFT amplitude convention	peak
FFT window correction	coherent_gain
FFT size	8,192
Window	hann
Averaging	linear

Measurement overview - FFT Spectrum Analysis / flight_daq.mpai [E09]

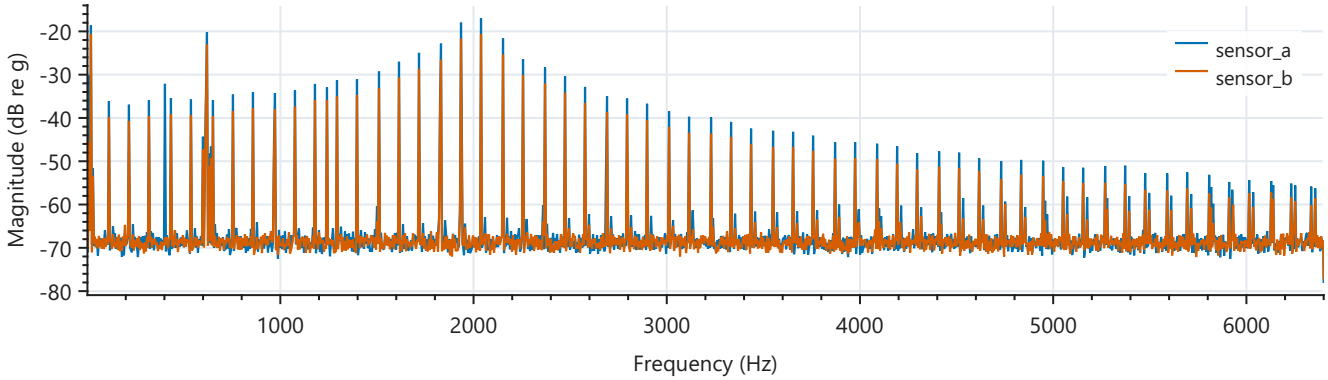


Figure 10: Source: FFT Spectrum Analysis · 2 channels · analysed 102,401 of 614,400 samples from sample 409,600 (Min-Max display reduction). dB amplitude from the cached native spectrum; all selected channels use the same amplitude convention.

Processing and operating conditions [E09]

Table 17: Processing and operating conditions [E09]

Parameter	Value
Sample rate (Hz)	12,800
FFT frequency resolution (Hz)	1.5625
FFT averages used	25
FFT amplitude convention	peak
FFT window correction	coherent_gain
FFT size	8,192
Window	hann
Averaging	linear

Measurement overview - Order Tracking / flight_daq.mpai / sensor_a [E10]

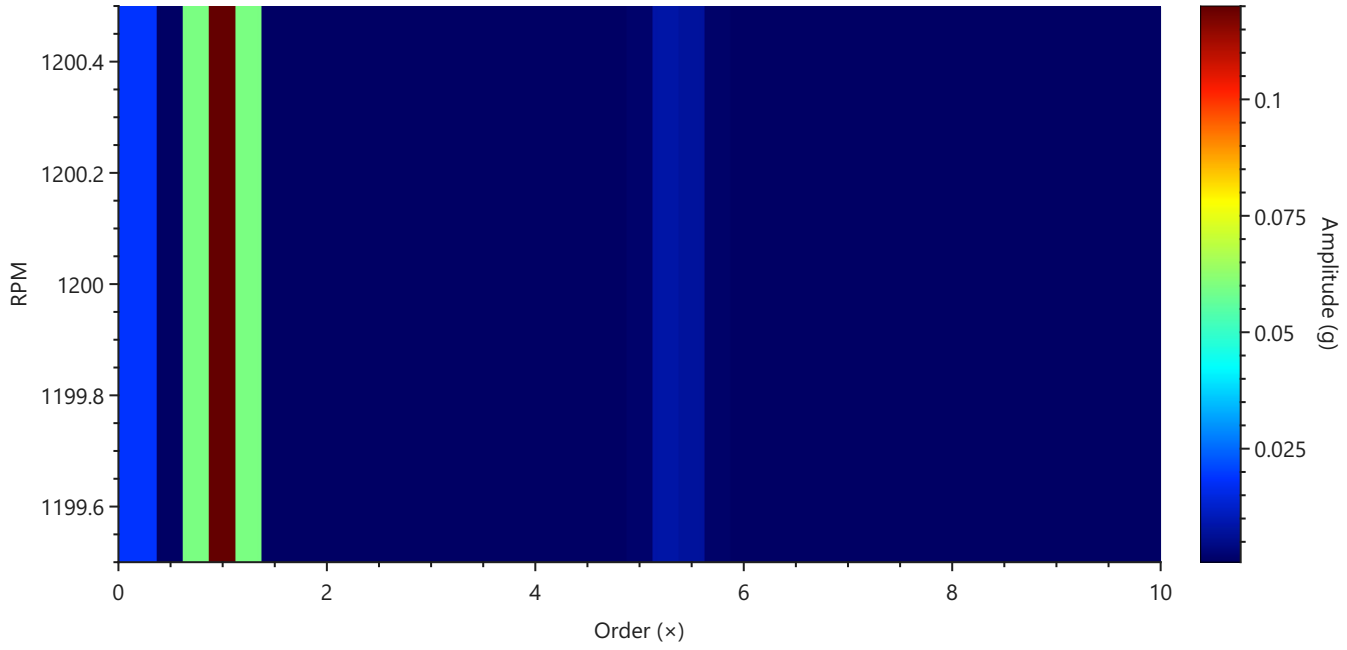


Figure 11: Source: Order Tracking

Processing and operating conditions [E10]

Table 18: Processing and operating conditions [E10]

Parameter	Value
Sample rate (Hz)	12,800
Window	hann

Selected numerical evidence - Relationship Explorer / mission_computer [E05]

Table 19: Selected numerical evidence - Relationship Explorer / mission_computer [E05]

Target	Channel	Pearson r	Valid pairs	Status
GBX_MONITOR_RMS	VHF_PTT	0.807543	960	Valid
GBX_MONITOR_RMS	GBX_VIB_HIGH	0.67434	960	Valid
GBX_MONITOR_RMS	OIL_TEMP_GBX	-0.204023	960	Valid
GBX_MONITOR_RMS	AUX_0396	0.163755	960	Valid
GBX_MONITOR_RMS	AUX_0126	0.16275	960	Valid
GBX_MONITOR_RMS	AUX_0351	0.162323	960	Valid
GBX_MONITOR_RMS	AUX_0036	0.161019	960	Valid
GBX_MONITOR_RMS	AUX_0261	0.160889	960	Valid

Display excerpt: rows 1-8 of 649. All rows remain in the connected result; analysis and acceptance checks use the complete result.

Selected engineering fields; other columns remain available in the connected result.

Relationship interpretation [E05]

Highest returned associations: VHF_PTT r=0.807543; GBX_VIB_HIGH r=0.67434. The coefficients compare paired samples within the controller source at the reported lag. They do not establish DAQ-to-controller synchronization or a physical cause. Similar coefficients for multiple channels do not constitute independent corroboration; check derived channels and collinearity. Small absolute coefficients provide little linear association in this selection; statistical significance and nonlinear effects are not established here.

Selected numerical evidence - Channel Profile / mission_computer [E06] [E27]

Table 20: Selected numerical evidence - Channel Profile / mission_computer [E06] [E27]

Channel	Unit	Samples	RMS	Crest factor	Excess kurtosis	Minimum	Maximum
GBX_MONITOR_RMS		160	0.197916	6.52613	48.3399	0.0832551	1.29162
NP_RPM		160	1,200	1	NaN	1,200	1,200
ENGINE_TORQUE		160	60.0067	1.00156	-1.36589	59.9	60.1

Selected engineering fields; other columns remain available in the connected result.

Selected numerical evidence - Channel Profile / mission_computer [E08] [E26]

Table 21: Selected numerical evidence - Channel Profile / mission_computer [E08] [E26]

Channel	Unit	Samples	RMS	Crest factor	Excess kurtosis	Minimum	Maximum
GBX_MONITOR_RMS		160	0.423474	3.75277	74.5248	0.35875	1.5892
NP_RPM		160	1,200	1	NaN	1,200	1,200
ENGINE_TORQUE		160	60.0186	1.00136	-1.41877	59.9	60.1

Selected engineering fields; other columns remain available in the connected result.

Selected numerical evidence - Channel Profile / flight_daq [E11]

Table 22: Selected numerical evidence - Channel Profile / flight_daq [E11]

Channel	Unit	Samples	RMS	Crest factor	Excess kurtosis	Minimum	Maximum
sensor_a	g	614,400	0.171174	6.82186	4.63631	-0.990556	1.16773
sensor_b	g	614,400	0.119389	6.45312	3.26223	-0.687852	0.770432

Selected engineering fields; other columns remain available in the connected result.

Channel comparison [E11]

sensor_a / sensor_b RMS ratio: 1.43375. This scalar ratio compares the recorded selection and units. Sensor roles and transfer paths must be checked before assigning a physical cause.

Selected numerical evidence - Channel Profile / ground_check [E12]

Table 23: Selected numerical evidence - Channel Profile / ground_check [E12]

Channel	Unit	Samples	RMS	Crest factor	Excess kurtosis	Minimum	Maximum
sensor_a	g	614,400	0.171149	6.83178	4.63725	-0.964351	1.16925
sensor_b	g	614,400	0.119371	6.41141	3.26911	-0.687273	0.765334

Selected engineering fields; other columns remain available in the connected result.

Channel comparison [E12]

sensor_a / sensor_b RMS ratio: 1.43376. This scalar ratio compares the recorded selection and units. Sensor roles and transfer paths must be checked before assigning a physical cause.

Selected numerical evidence - Channel Profile / mission_computer [E21] [E25]

Table 24: Selected numerical evidence - Channel Profile / mission_computer [E21] [E25]

Channel	Unit	Samples	RMS	Crest factor	Excess kurtosis	Minimum	Maximum
GBX_MONITOR_RMS		960	0.342855	4.64458	15.4222	0.0832551	1.59242
NP_RPM		960	1,200	1	NaN	1,200	1,200
ENGINE_TORQUE		960	60.0035	1.00161	-1.48221	59.9	60.1

Selected engineering fields; other columns remain available in the connected result.

5. Engineering interpretation

Interpretation

The independent checks support a provisional outer-race hypothesis and contradict a primary-sensor installation-only explanation. They do not establish a confirmed physical cause or exclude every installation issue.

Outer-race bearing fault - supported

The evidence supports a BPFO (Outer Race Fault) in the rolling bearing on the driving pinion shaft, based on 5 matched harmonic components detected in both flight and ground check recordings.

Supporting evidence: [E17], [E18], [E19], [E20]

Counterevidence: None recorded

Unclassified explanation requiring a discriminating test - rejected

A primary-sensor installation problem alone explains the sustained vibration.

Supporting evidence: None recorded

Counterevidence: [E17], [E18], [E19], [E20]

Reason: Independent witness channels and the separately recorded primary-remounted ground check contradict this installation-only explanation, without excluding every sensor issue.

6. Uncertainty and follow-up

Uncertainty

Physical damage and repair effectiveness remain unconfirmed.

Recommended checks

Confirm bearing condition by independent physical inspection and a matched post-intervention repeat. No component repair or operational release is verified.

Required follow-up and limitations

- physical bearing inspection
- post-intervention repeat verification
- Supported hypotheses are provisional; physical cause and repair effectiveness require independent verification.
- Rejection concerns the tested explanation in recorded conditions; it does not rule out every physical defect.

7. Technical appendix - evidence register

Evidence references below identify the recorded computations and their source scope.

Report identity

Created: 2026-11-12T23:34:05+00:00
Status: Draft engineering assessment
Author: Application-assisted investigation; analyst/model identity not supplied.
Software: MachinePulseAI; build identifier not supplied

Computation receipts

Table 25: Computation receipts

Evidence	Source	Check	Channels	Status	Selection
[E01]	[S01]	Reference lookup		ok	Not recorded / reference only
[E02] [E23]	[S02]	Analysis	Channel list in JSON	Earlier result; latest is [E28]	960 samples from sample 0
[E03]	[S03]	Analysis	Channel list in JSON	Earlier result; latest is [E24]	614,400 samples from sample 0
[E04]	[S04]	Analysis	Channel list in JSON	Earlier result; latest is [E22]	614,400 samples from sample 0
[E05]	[S02]	Analysis	Channel list in JSON	Latest result	960 samples from sample 0
[E06]	[S02]	Analysis	GBX_MONITOR_RMS, NP_RPM, ENGINE_TORQUE	Earlier result; latest is [E27]	160 samples from sample 39
[E07]	[S03]	Analysis	sensor_a, sensor_b	Latest result	102,401 samples from sample 25,600
[E08]	[S02]	Analysis	GBX_MONITOR_RMS, NP_RPM, ENGINE_TORQUE	Earlier result; latest is [E26]	160 samples from sample 639
[E09]	[S03]	Analysis	sensor_a, sensor_b	Latest result	102,401 samples from sample 409,600
[E10]	[S03]	Analysis	shaft_rpm, keyphasor, sensor_a, ['sensor_a']	Latest result	614,400 samples from sample 0
[E11]	[S03]	Analysis	sensor_a, sensor_b	Latest result	614,400 samples from sample 0
[E12]	[S04]	Analysis	sensor_a, sensor_b	Latest result	614,400 samples from sample 0
[E13]	[S03]	Analysis	shaft_rpm, sensor_a	Latest result	614,400 samples from sample 0
[E14]	[S03]	Analysis	shaft_rpm, sensor_b	Latest result	614,400 samples from sample 0
[E15]	[S04]	Analysis	shaft_rpm, sensor_a	Latest result	614,400 samples from sample 0

Evidence	Source	Check	Channels	Status	Selection
[E16]	[S04]	Analysis	shaft_rpm, sensor_b	Latest result	614,400 samples from sample 0
[E17]	[S03]	BPFO	shaft_rpm, sensor_a	supported	614,400 samples from sample 0
[E18]	[S03]	BPFO	shaft_rpm, sensor_b	supported	614,400 samples from sample 0
[E19]	[S04]	BPFO	shaft_rpm, sensor_a	supported	614,400 samples from sample 0
[E20]	[S04]	BPFO	shaft_rpm, sensor_b	supported	614,400 samples from sample 0
[E21]	[S02]	Analysis	GBX_MONITOR_RMS, NP_RPM, ENGINE_TORQUE	Earlier result; latest is [E25]	960 samples from sample 0
[E22]	[S04]	Analysis	Channel list in JSON	Latest result	614,400 samples from sample 0
[E24]	[S03]	Analysis	Channel list in JSON	Latest result	614,400 samples from sample 0
[E25]	[S02]	Analysis	GBX_MONITOR_RMS, NP_RPM, ENGINE_TORQUE	Latest result	960 samples from sample 0
[E26]	[S02]	Analysis	GBX_MONITOR_RMS, NP_RPM, ENGINE_TORQUE	Latest result	160 samples from sample 639
[E27]	[S02]	Analysis	GBX_MONITOR_RMS, NP_RPM, ENGINE_TORQUE	Latest result	160 samples from sample 39
[E28]	[S02]	Analysis	Channel list in JSON	Latest result	960 samples from sample 0
[E29]	[S05]	Analysis		Latest result	Not recorded / reference only
[E30]	[S06]	Analysis		Latest result	Not recorded / reference only

Full evidence identities, source paths and processing receipts are retained in the exported .evidence.json companion.

Source register

Table 26: Source register

Source	Recording	Analysed file state
S01	Reference material	Not recorded
S02	mission_computer.csv	mission_computer_19d87ceeb004.mpai: 5.46 MB, modified 2026-11-13 01:56:41 +0300
S03	flight_daq.mpai	127.76 MB, modified 2026-11-13 01:56:34 +0300
S04	ground_check.mpai	127.76 MB, modified 2026-11-13 01:56:34 +0300
S05	Reference material	Not recorded
S06	Reference material	Not recorded

Applicable reference register

Table 27: Applicable reference register

Reference	Document	Revision	Evidence
AIR-QUALITY-R2	quality.md	R2	[E01]
machine.json	machine.json	R2	[E01]
AIR-SETUP-R2	setup.md	R2	[E01]
quality.md	quality.md	R2	[E01]
setup.md	setup.md	R2	[E01]